

CS 486/686

Heuristic Search

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Lecture 3

RN 3.5 · PM 3.6–3.7

Search ›

Uncertainty ›

Decisions ›

Learning

Learning goals

- Motivate heuristic search
- Trace LCFS, GBFS, A*
- Analyze: space, time, completeness, optimality
- Design and dominate admissible heuristics
- Prune the search space

Which state is closer to the goal?

State A

5	3	
8	7	6
2	4	1

State B

1	2	3
4	5	
7	8	6

Goal

1	2	3
4	5	6
7	8	

B is just one move away. Humans see this instantly — uninformed search does not.

Uninformed vs heuristic

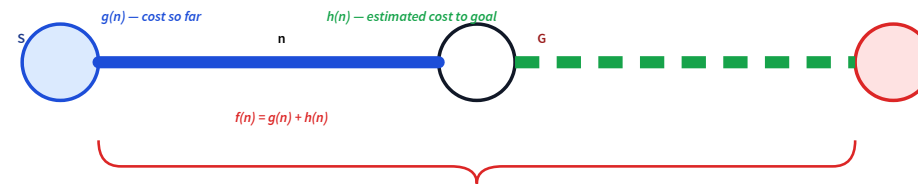
Uninformed

- Treats every state the same
- No idea which is closer to goal

Heuristic

- Estimates distance to goal
- Expands the most promising state

Two quantities at every node n



Solid **blue**: known. Dashed **green**: estimated. **Red**: total estimate.

What makes a good heuristic?



**Problem-
specific**



Non-negative



$h(\text{goal}) = 0$



**Cheap to
compute**

Three algorithms, one knob

Frontier is a priority queue. The three algorithms differ in the priority key.

LCFS

$$g(n)$$

cheapest-path-so-far

GBFS

$$h(n)$$

looks-closest-to-goal

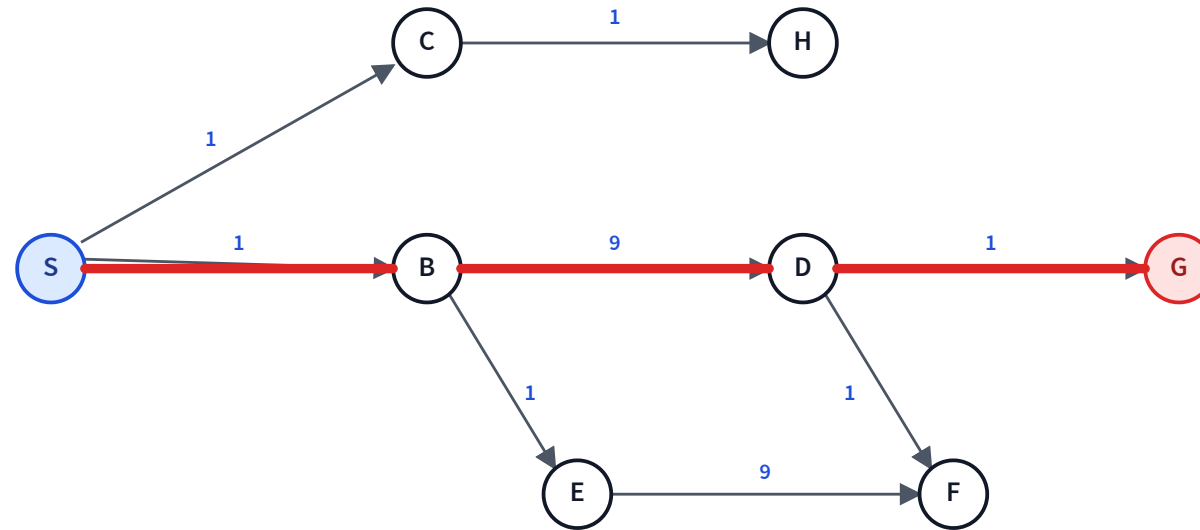
A*

$$g(n) + h(n)$$

total estimated cost

The trace graph

All four traces use this graph. Edge labels are **costs**. Note that F has two parents (E and D).



Optimal path: S → B → D → G with total cost **11**.

Lowest-cost-first search (LCFS)

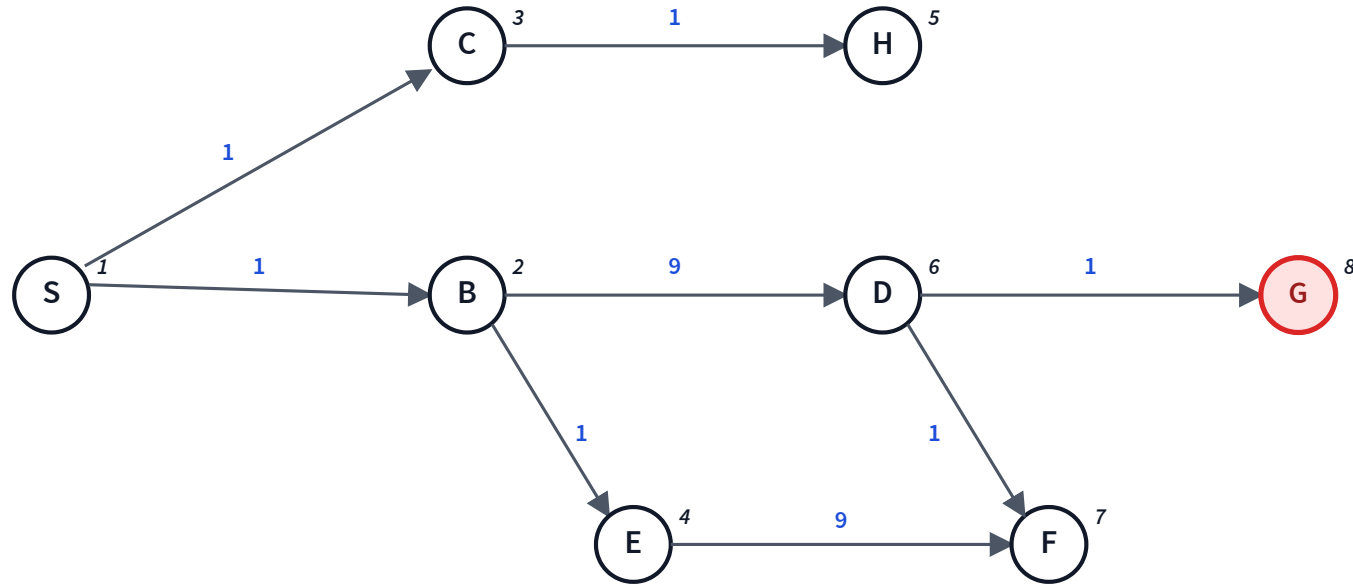
- Frontier = **priority queue**, key = $g(n)$
- Pop the cheapest path so far

a.k.a. **Uniform-cost search (UCS)** in RN, or **Dijkstra's algorithm** in algorithms textbooks.

Trace: LCFS

Path notation SBE: 2 = path $S \rightarrow B \rightarrow E$ with cumulative cost 2. Ties broken alphabetically.

Frontier:



LCFS — properties

Space	$O(b^d)$ <i>exponential</i>
Time	$O(b^d)$ <i>exponential</i>
Complete?	Yes <i>finite branching, edge cost $\geq \varepsilon > 0$</i>
Optimal?	Yes <i>under the same conditions</i>

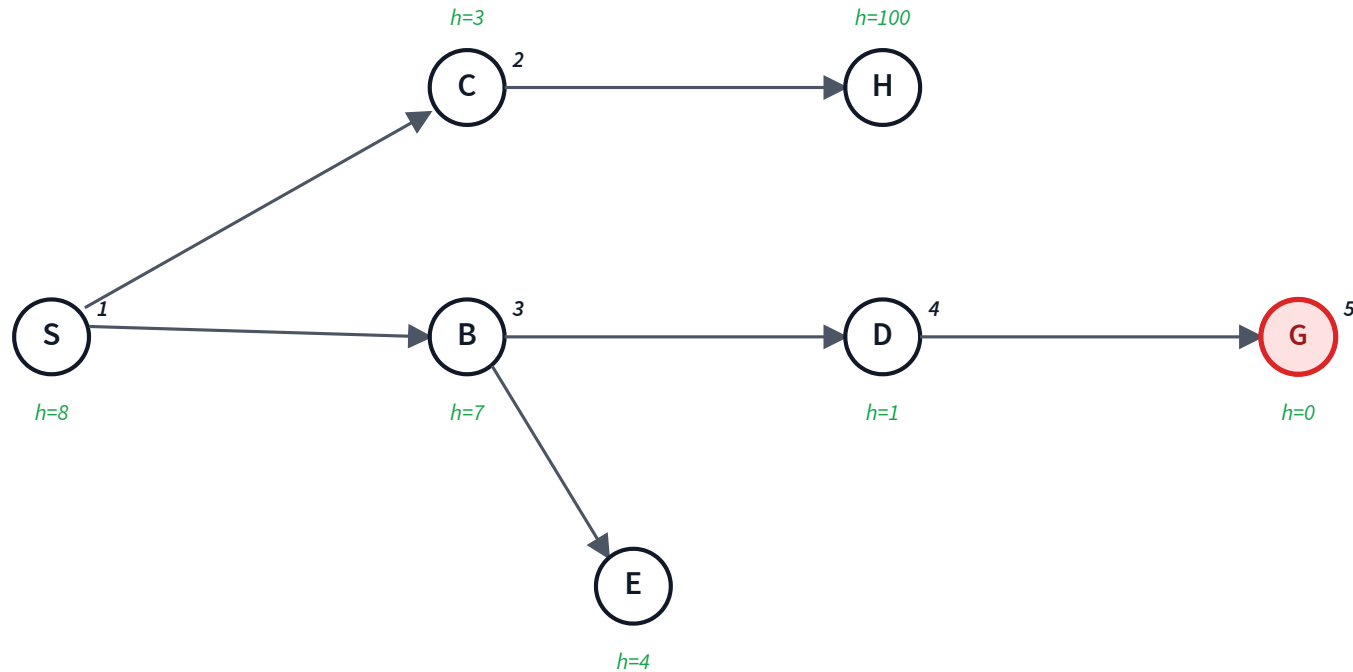
Greedy best-first search (GBFS)

- Frontier = **priority queue**, key = $h(n)$
- Pop the most promising path (lowest h)

Trace: GBFS

Path notation SBD: 1 = path with $h(D) = 1$. Heuristic values shown in green.

Frontier:



GBFS — properties

Space	$O(b^m)$ <i>exponential</i>
Time	$O(b^m)$ <i>exponential</i>
Complete?	No <i>can get stuck following a misleading heuristic</i>
Optimal?	No <i>greedy choice may be wrong</i>

Intuition: what each algorithm prioritises

	LCFS	GBFS
Frontier sorted by	$g(n)$ — cost so far	$h(n)$ — estimated cost to goal
Closest uninformed cousin	BFS with weights. When every edge costs 1, LCFS = BFS exactly.	DFS guided by h . Chases one direction aggressively.
Search shape	A cost-wavefront that grows <i>outward</i> from the start.	A narrow probe that dives toward whatever <i>looks</i> closest.

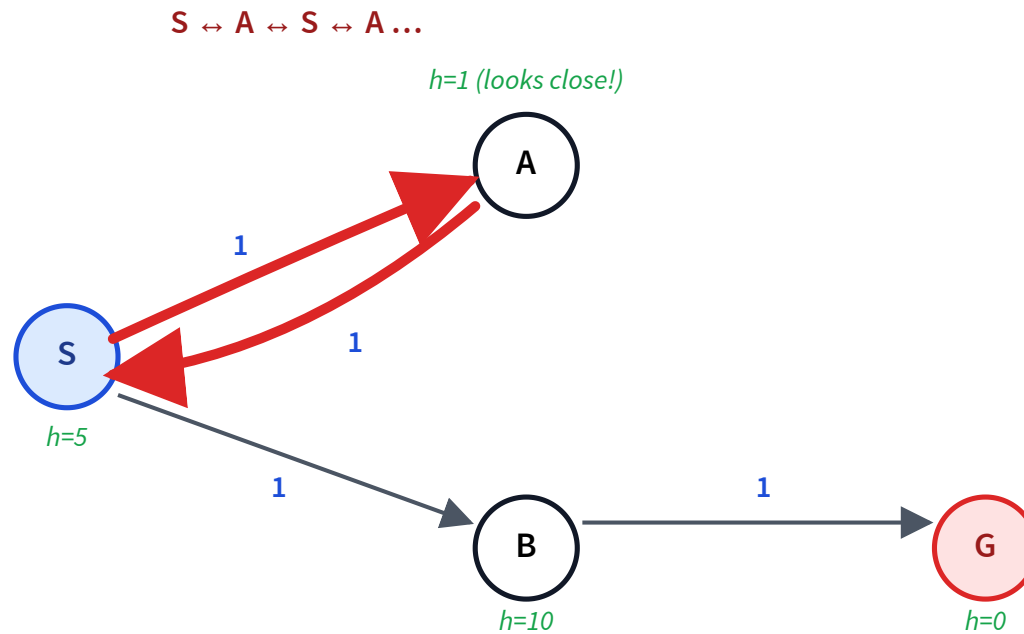
Intuition: what guarantees follow

	LCFS	GBFS
Complete?	Yes — wavefront eventually reaches every reachable state.	No — low- h cycles can trap it.
Optimal?	Yes — first pop of G has the smallest g .	No — h alone can prefer a costly “looks-close” detour.

Worst-case time / space: LCFS is $O(b^d)$; GBFS is $O(b^m)$.

GBFS — not complete

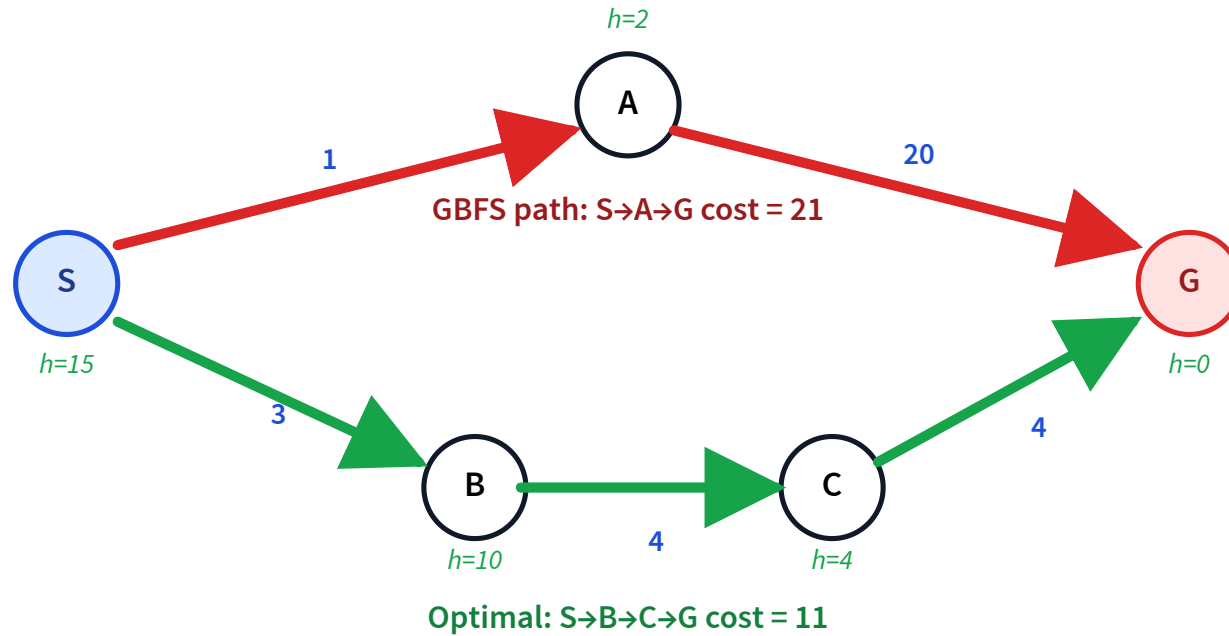
A low- h cycle can trap GBFS forever — it never explores the path that actually reaches G.



Pop order: SA ($h=1$), SAS ($h=5$), SASA ($h=1$), ... SB ($h=10$) is always outranked — G never reached.

GBFS — not optimal

Even when GBFS finds the goal, the path can be far from cheapest.



A* search

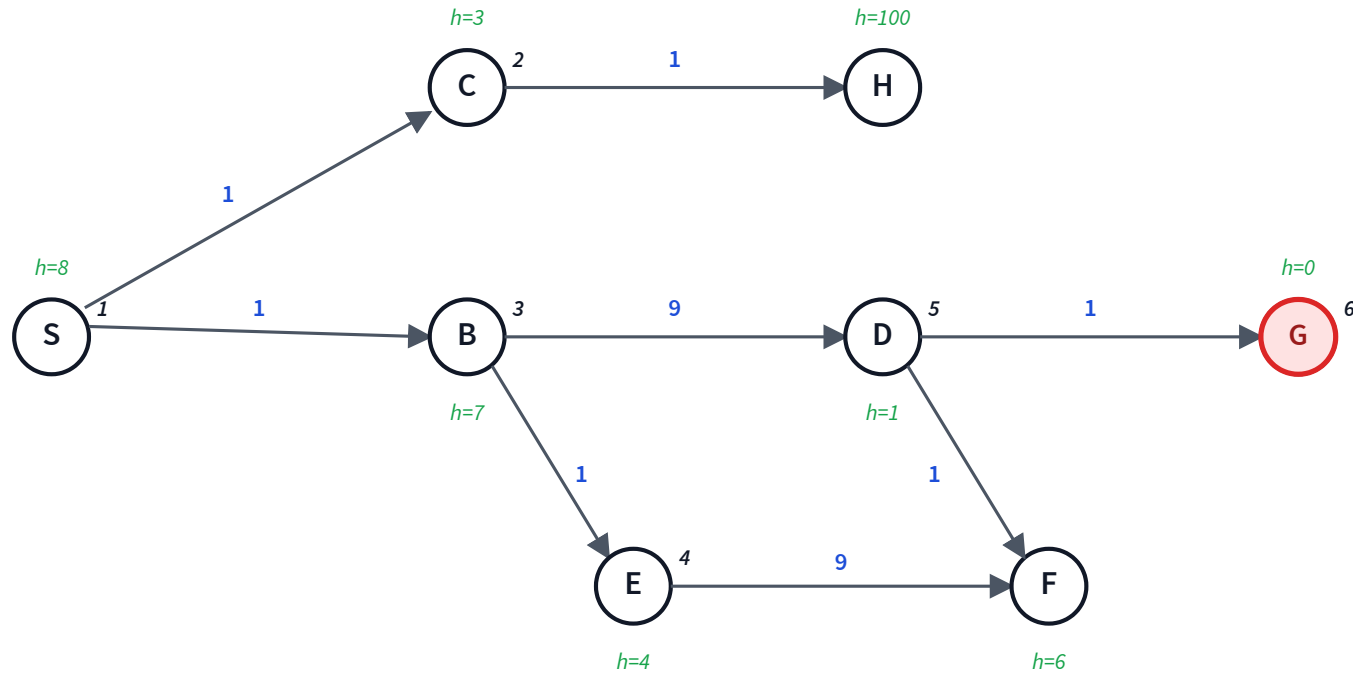
- Frontier = **priority queue**, key = $f(n) = g(n) + h(n)$
- Pop the path with smallest total estimated cost

A* combines LCFS's **cost-so-far** with GBFS's **goal-pull**.

Trace: A*

Path notation SBE: 6 = path with $f = g + h = 6$. **g** in blue, **h** in green.

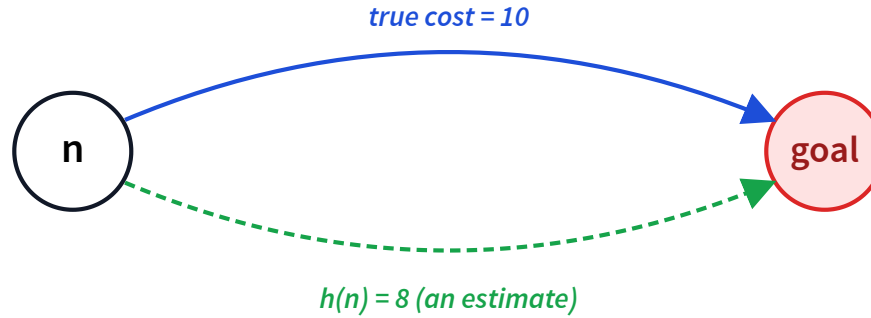
Frontier:



A* — properties

Space	$O(b^m)$ <i>worst case; often much better</i>
Time	$O(b^m)$ <i>worst case; often much better</i>
Complete?	Yes <i>under the usual finite-branching condition</i>
Optimal?	Yes <i>if h is admissible</i>

Admissible heuristic



Admissible: $h(n) \leq$ true cost from n to any goal, for every n . *Never over-estimates.*

Theorem. If h is admissible, A^* returns an optimal path.

A* optimality — proof sketch

Assume A* returns cost C^* , but the true optimum is $C' < C^*$.

1. Some node N on the optimal path is still on the frontier at termination.
2. Admissibility $\Rightarrow f(N) = g(N) + h(N) \leq C'$.
3. So $f(N) \leq C' < C^*$ — A* would have popped N first. **Contradiction.** ■

A^* is optimally efficient

No other optimal algorithm using the same h expands fewer nodes.

Manhattan distance

Sum of grid distances of each tile to its goal position.

Initial			Goal					
5	3		1	2	3			
8	7	6	4	5	6			
2	4	1	7	8				
tile	1	2	3	4	5	6	7	8
distance	4	3	1	2	2	0	2	2

$$\text{Manhattan}(\text{start}) = 4 + 3 + 1 + 2 + 2 + 0 + 2 + 2 = 16$$

Misplaced tile count

Number of tiles not yet in their goal position.

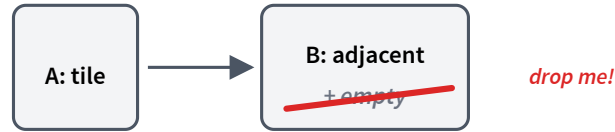
Initial			Goal		
5	3		1	2	3
8	7	6	4	5	6
2	4	1	7	8	

Red = misplaced. Green = already at goal.

Misplaced(start) = 7

Both heuristics are **admissible**.

Recipe: relax the problem



1. **Relax** the problem by dropping constraints.
2. **Solve** the relaxed problem (without search).
3. That cost is an **admissible heuristic** for the original.

An easier problem gives a lower bound — safe to use as an estimate.

Quick check: heuristic from relaxation

Q. Drop the "B is empty" constraint: a tile can move from A to B if A and B are adjacent. Which heuristic does this give?

- A. Manhattan distance
- B. Misplaced tile count
- C. Another heuristic not listed

A — Manhattan distance. Without the "B must be empty" constraint, each tile slides freely along the grid to its goal — cost = number of grid steps = Manhattan distance.

Dominating heuristics

Definition. h_2 dominates h_1 if $h_2(n) \geq h_1(n)$ for all n , and $h_2(n) > h_1(n)$ for some n .

Theorem. If h_2 dominates h_1 , A^* with h_2 expands no more nodes than A^* with h_1 . Higher = better — as long as still admissible.

Manhattan vs Misplaced

Q. On our example state, Manhattan = 16 and Misplaced = 7. Which heuristic dominates the other in general?

- A. Manhattan dominates Misplaced
- B. Misplaced dominates Manhattan
- C. Neither dominates the other

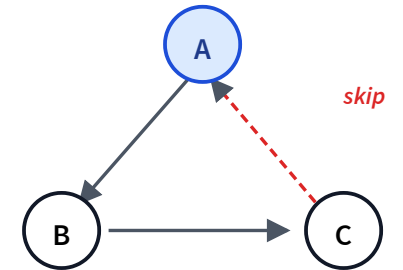
A — Manhattan dominates. Each misplaced tile contributes at least 1 to both heuristics; Manhattan adds the actual grid distance, so $\text{Manhattan} \geq \text{Misplaced}$, and strictly $>$ whenever any tile is more than one step from its goal.

Cycle pruning

Don't follow a path that revisits a node already on the current path.

When expanding path $\langle n_0, \dots, n_k \rangle$:

1. For each neighbour n of n_k :
2. If n already appears in $\langle n_0, \dots, n_k \rangle \rightarrow$ **skip**.
3. Else add $\langle n_0, \dots, n_k, n \rangle$ to the frontier.



Check is linear in path length. Saves DFS from infinite loops.

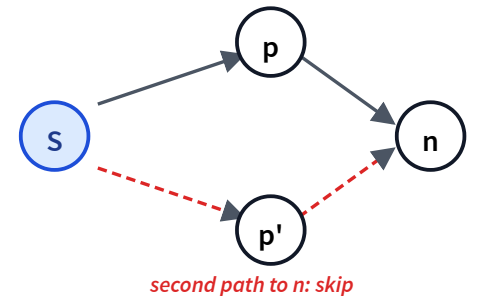
Multi-path pruning

Don't expand a node we've already expanded (across *different* paths). Cycle pruning is a special case.

Maintain an explored set, initialized to $\{ \}$.

Pop path $\langle n_0, \dots, n_k \rangle$. Then:

1. If $n_k \in \text{explored}$ $\rightarrow$ skip.
2. Add n_k to **explored**.
3. If n_k is the goal $\rightarrow$ return the path.
4. Else push each neighbour-extension to the frontier.



Caveats of multi-path pruning

- Repeats still pushed — skipped on pop
- Trades memory (explored set) for time
- May sacrifice optimality (see next slides)

LCFS + multi-path pruning

Q. LCFS with multi-path pruning — can it discard the optimal solution?

A. Yes, it can

B. No, never

B — No. First pop of any node already has the smallest g ; later paths to it can only be costlier.
Safe to prune.

A* + multi-path pruning

Q. A* with admissible h + multi-path pruning — can it discard the optimal?

A. Yes, it can

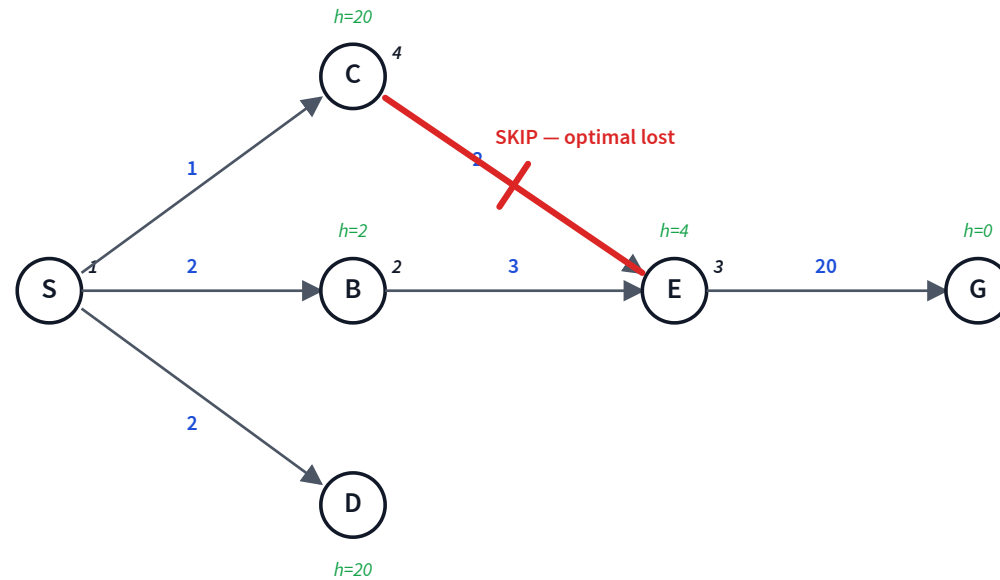
B. No, never

A — Yes, it can. A* orders by $f = g + h$, so the first pop of n need not have the smallest g . A cheaper $g(n)$ found later is then discarded. *Counterexample next.*

A* + multi-path: counterexample

Optimal path is S→C→E→G (cost 23). With multi-path pruning, A* returns S→B→E→G (cost 25) instead.

Frontier & Explored:



Consistent (monotone) heuristic

Definition. h is consistent if for every edge $m \rightarrow n$:

$$h(m) \leq \text{cost}(m, n) + h(n)$$

i.e., the heuristic decreases by no more than the edge cost as we move toward the goal.

Theorem. If h is consistent, A* with multi-path pruning is **optimal**.

Most natural heuristics (Manhattan, straight-line distance, ...) are consistent.

Where A^* can mis-prune

Why does **LCFS** need no extra condition, but A^* does?

LCFS — sorts by $g(n)$:

- first pop of n already has the smallest $g(n)$ — safe.

A^* — sorts by $f = g + h$:

- $h(n)$ cancels at n , but differs across prefixes p, p' ;
- a big h -gap can pop the larger- g path first $\rightarrow n$ committed to a worse prefix.

How consistency restores the guarantee

Bound h -change along every edge $p \rightarrow n$:

$$h(p) - h(n) \leq c(p, n) \Leftrightarrow h(p) \leq c(p, n) + h(n)$$

— that's exactly **consistency**.

Chain of consequences:

- f is non-decreasing along every path
- first pop of n has the smallest g — LCFS guarantee, restored
- $\therefore$ multi-path pruning is safe

Five search algorithms

Algorithm	Frontier key	Space	Time	Complete?	Optimal?
DFS	LIFO (stack)	$O(bm)$	$O(b^m)$	No <i>cycles</i>	No
BFS	FIFO (queue)	$O(b^d)$	$O(b^d)$	Yes	Yes <i>unit cost</i>
LCFS	$g(n)$	$O(b^d)$	$O(b^d)$	Yes	Yes
GBFS	$h(n)$	$O(b^m)$	$O(b^m)$	No	No
A*	$g(n) + h(n)$	$O(b^m)$	$O(b^m)$	Yes	Yes <i>admissible h</i>

Learning goals (recap)

- ✓ Motivate heuristic search
- ✓ Trace LCFS, GBFS, A*
- ✓ Analyze space, time, completeness, optimality
- ✓ Design and dominate admissible heuristics
- ✓ Prune the search space (cycle, multi-path, consistency)